

Housing Recovery and Reconstruction Platform
(HRRP)

15 November, 2016

***Stabilizing Weak Masonry for
Schools and Houses:***

*An approach to Rubble Masonry in Mud
Mortar Construction in Earthquake Areas*

By

Randolph Langenbach



Bam, IRAN earthquake, 2002



Bam, IRAN earthquake, 2002

- Destruction in just 12 Seconds
- Buildings that were untouched in 150 years survived.









- Destruction in just 12 Seconds
- Buildings that were untouched in 150 years survived.
- The thicker the wall, the more likely it collapsed.



- TERMITES? !







The 2015 earthquakes in Nepal



Nepal, 2015



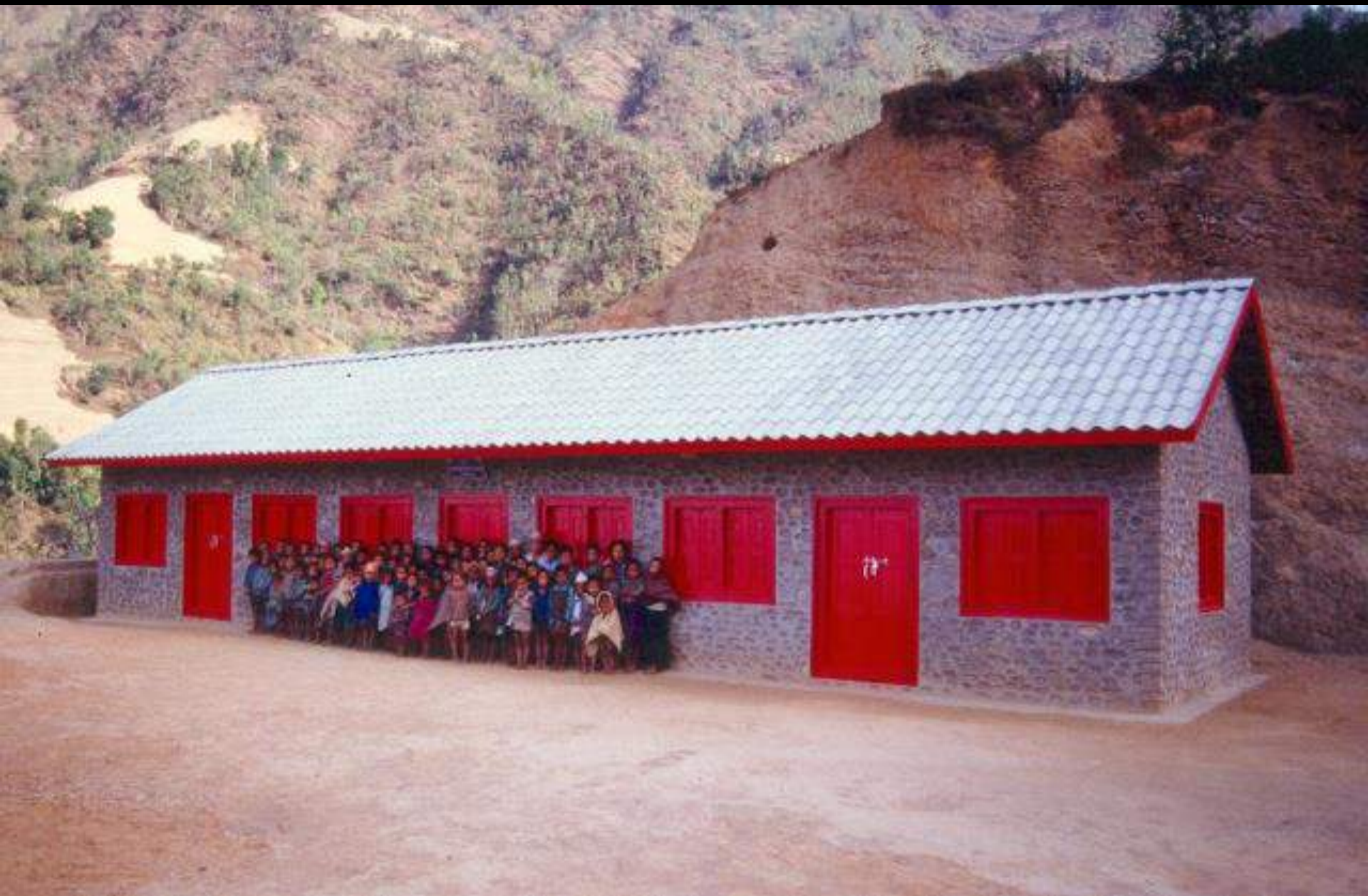
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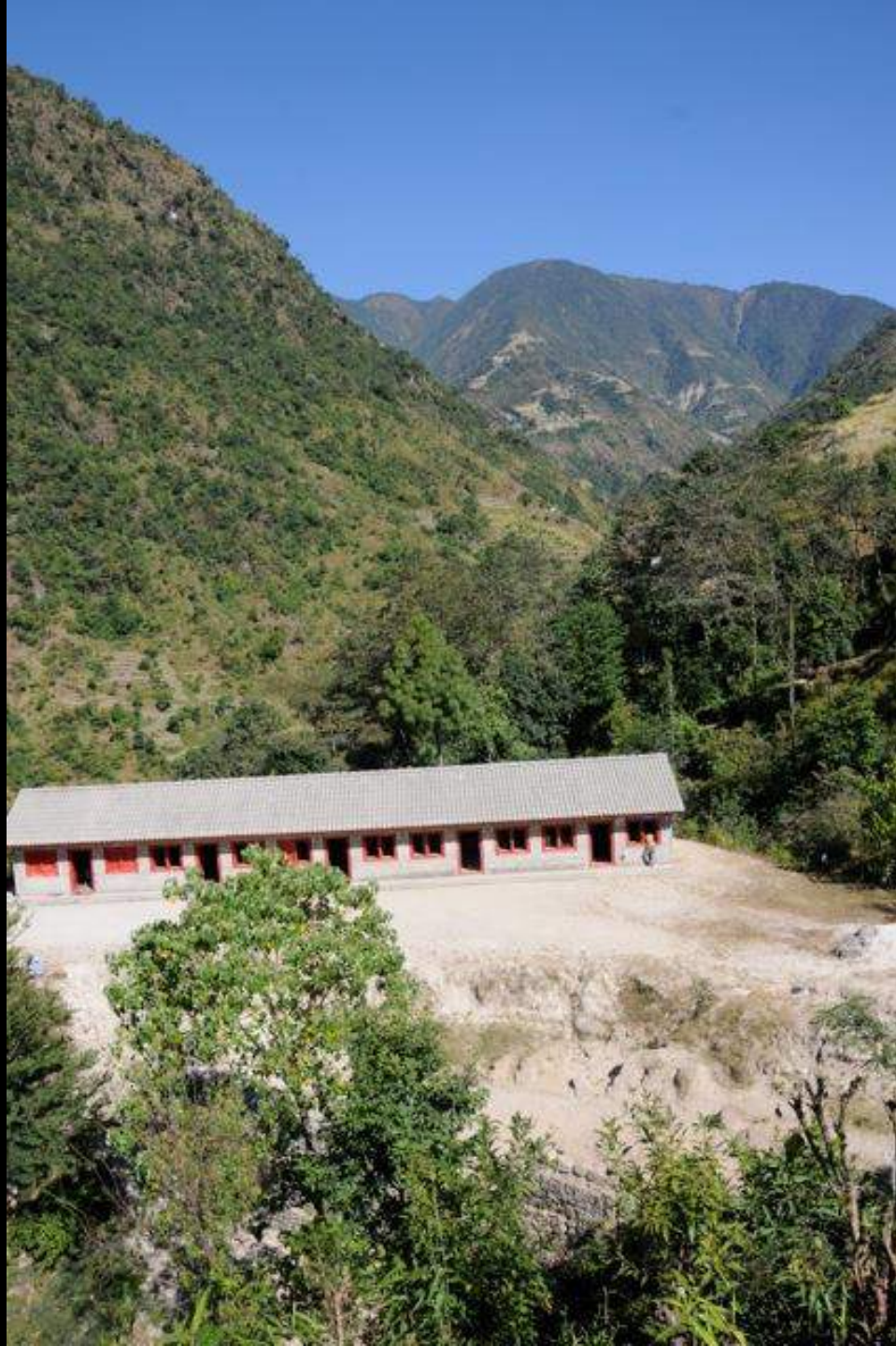


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School with RC
Bands,
Sangachok,
in Sindupalchok







**Amatrice, Italy after 24
August 2016 earthquake**



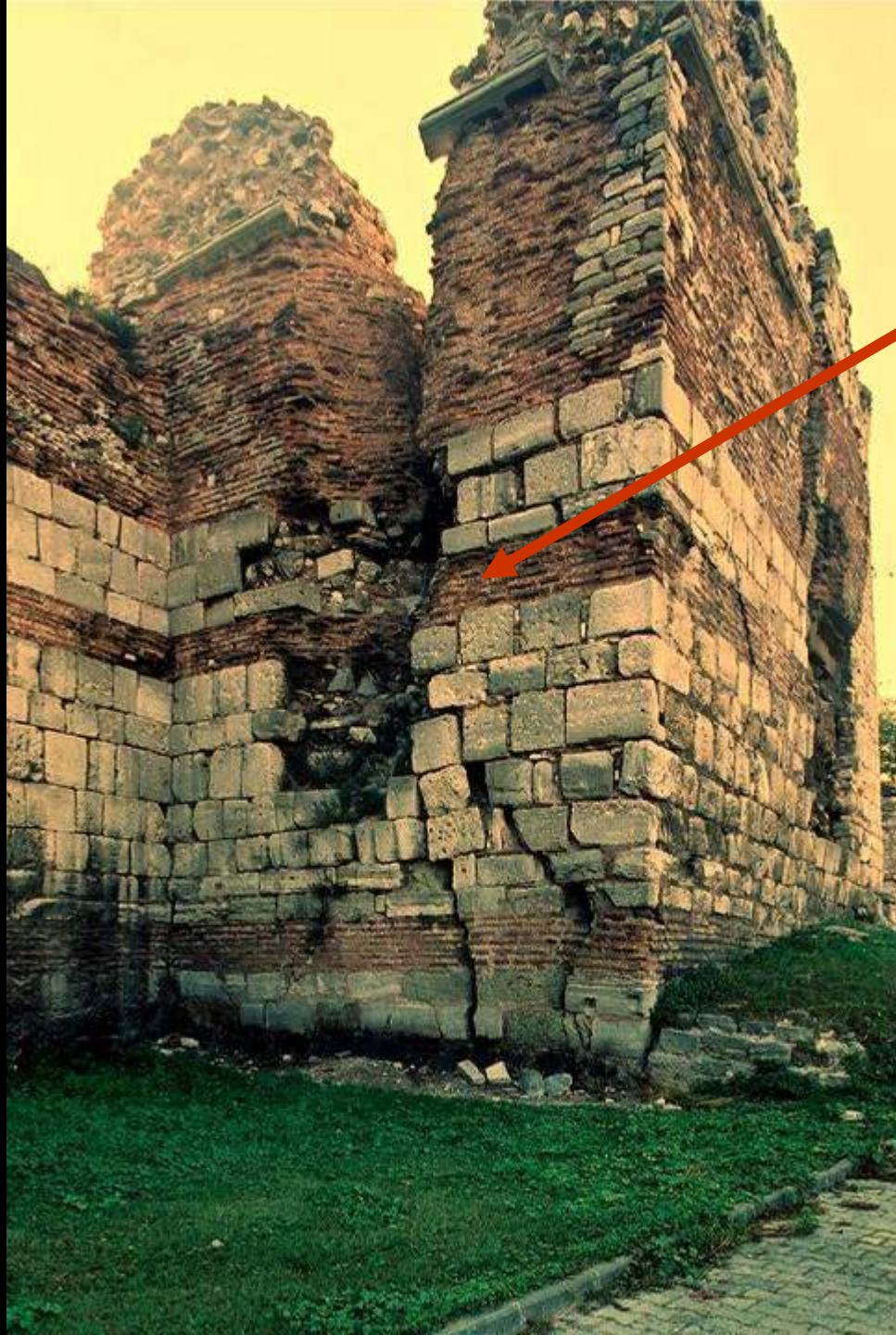




**Amatrice, Italy after 24
August 2016 earthquake**



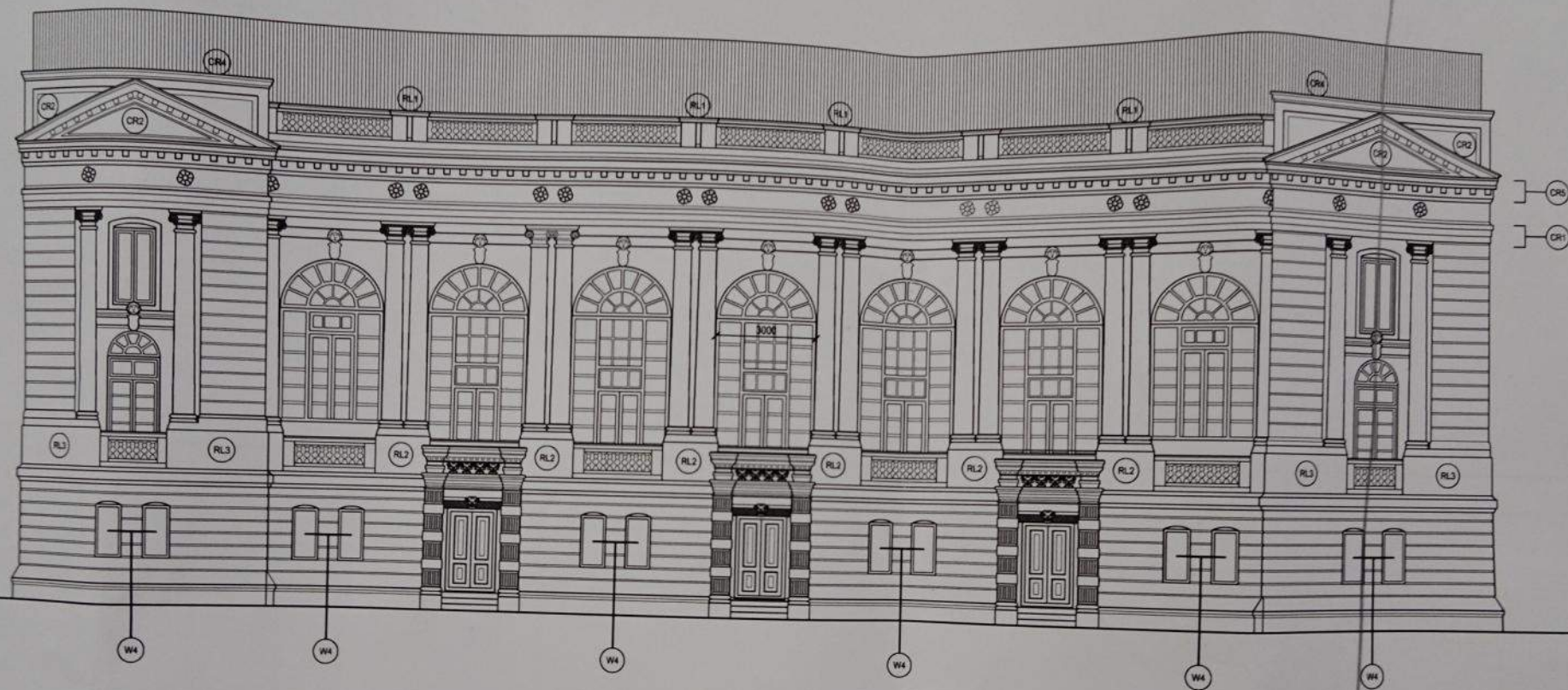




HATIL – a term used for the horizontal bands in masonry either of wood or of masonry (usually brick).







FRONT(SOUTH) ELEVATION

1:100

SRINAGAR, KASHMIR







dhajji dewari

**“patchquilt wall” in old
Persian**



dhajji dewari

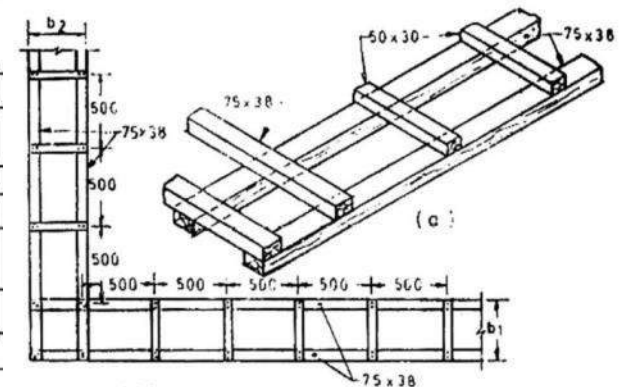
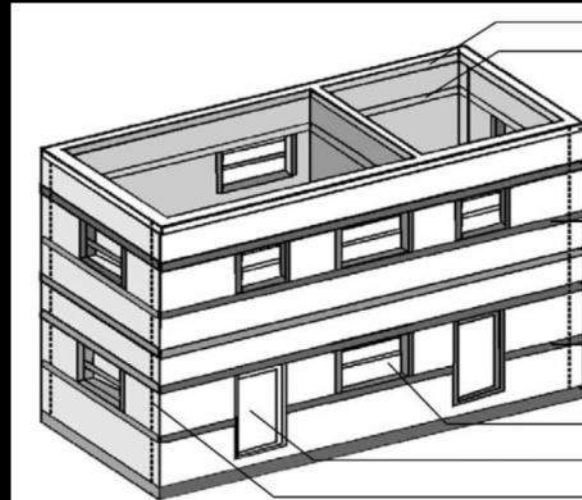
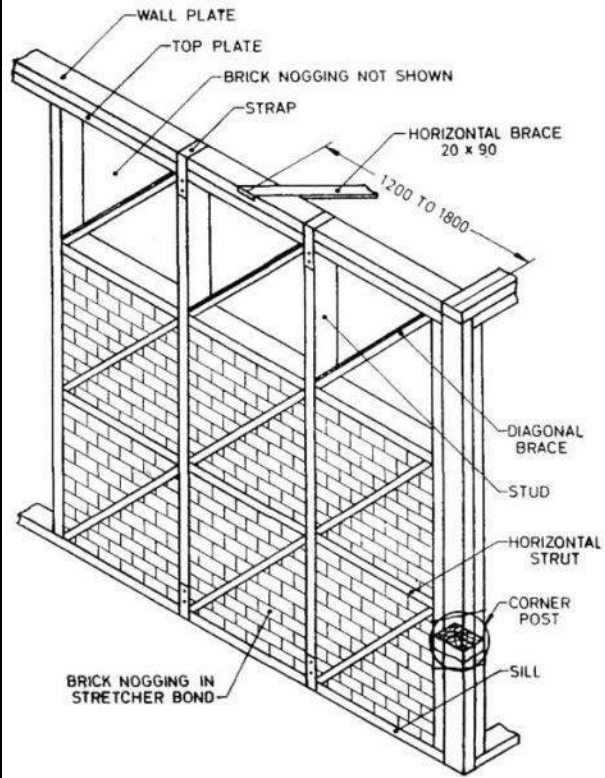
“patchquilt wall” in old Persian

Taq

(in Pakistan: *bhatar*)

Timber-laced masonry
bearing wall
construction

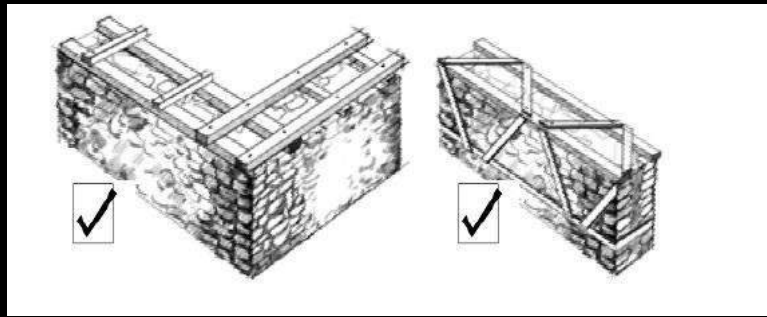




Nepal

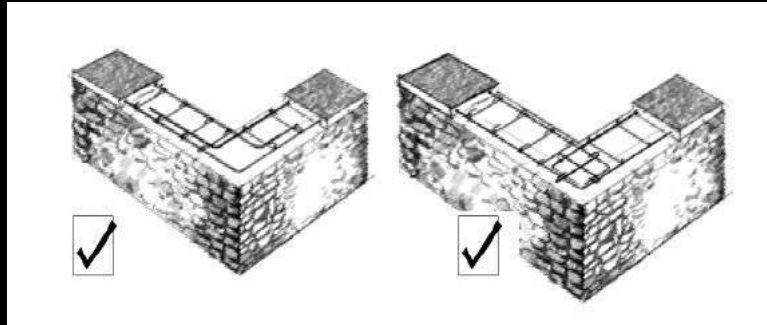


Nepal, 2015



Timber Bands?

...But there is a shortage of Timber?



Then how about Concrete?



...hand carried
sometimes miles
from the nearest
road???

*You have got to
be kidding!*

Then what else is possible?

What about - - - - “GABION BANDS”

Inspired by using a wire mesh that is
commonly available for fences?

...or maybe better – polypropylene geo-grids.

‘Bands’ (ring beams)



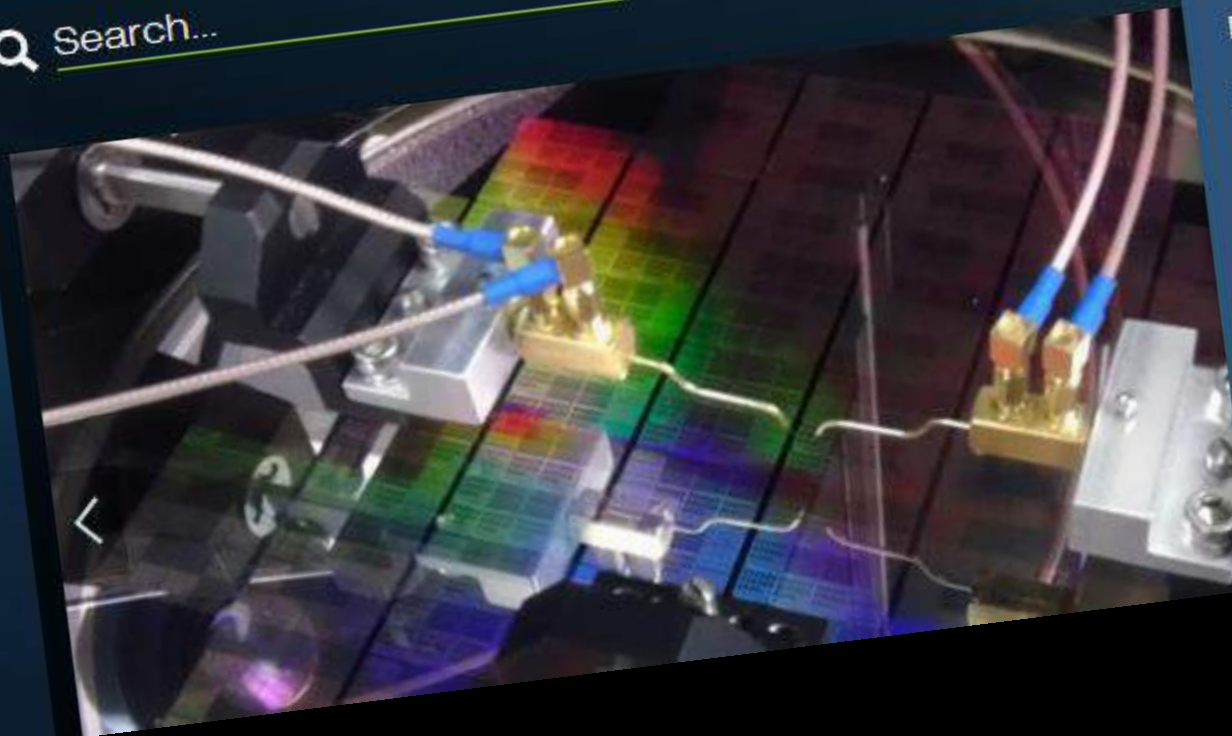
not ‘baskets’



PBS

NOVA

Search...



TOPICS

- ANCIENT WORLDS
- BODY + BRAIN
- EVOLUTION
- MILITARY + ESPIONAGE
- NATURE
- PHYSICS + MATH
- PLANET EARTH
- SPACE + FLIGHT
- TECH + ENGINEERING





Sunar Family Farm, 2015



Sunar Family Farm, 2015

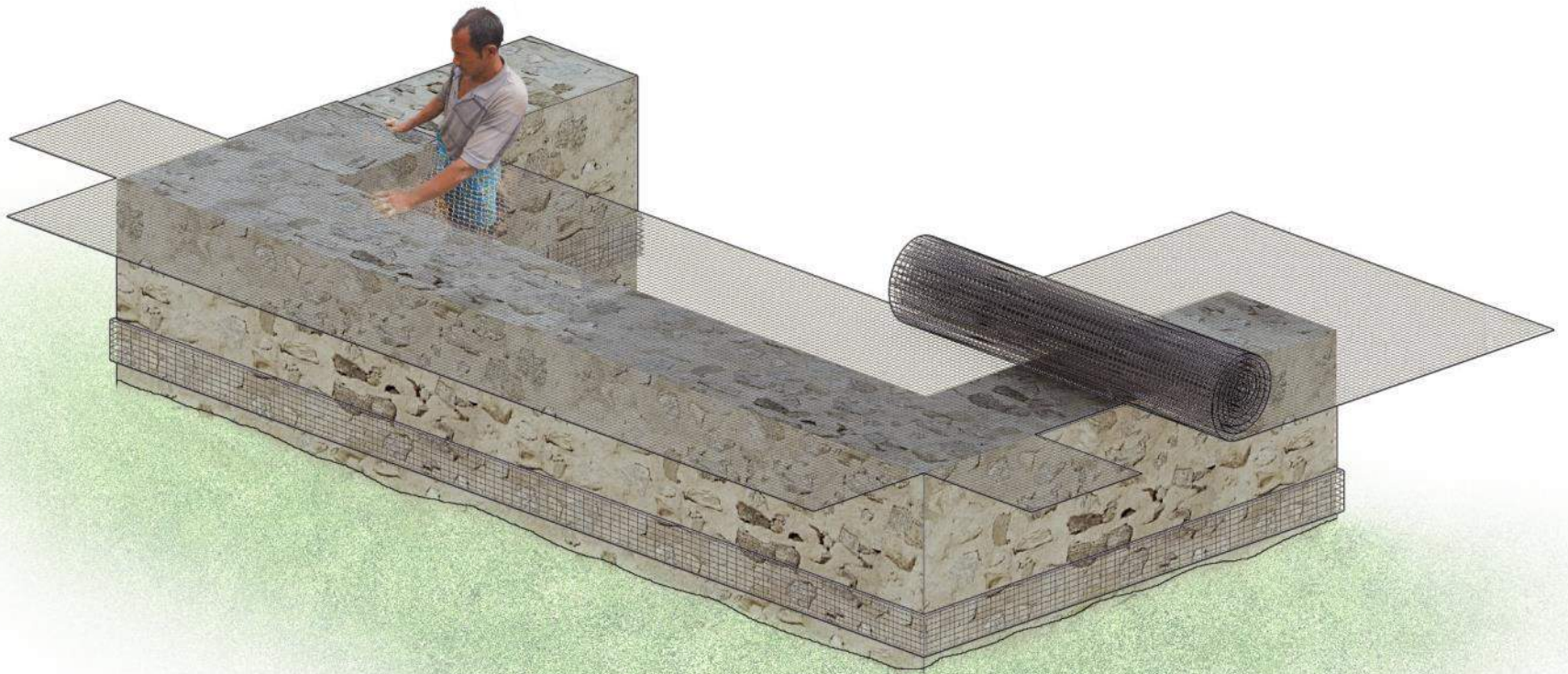


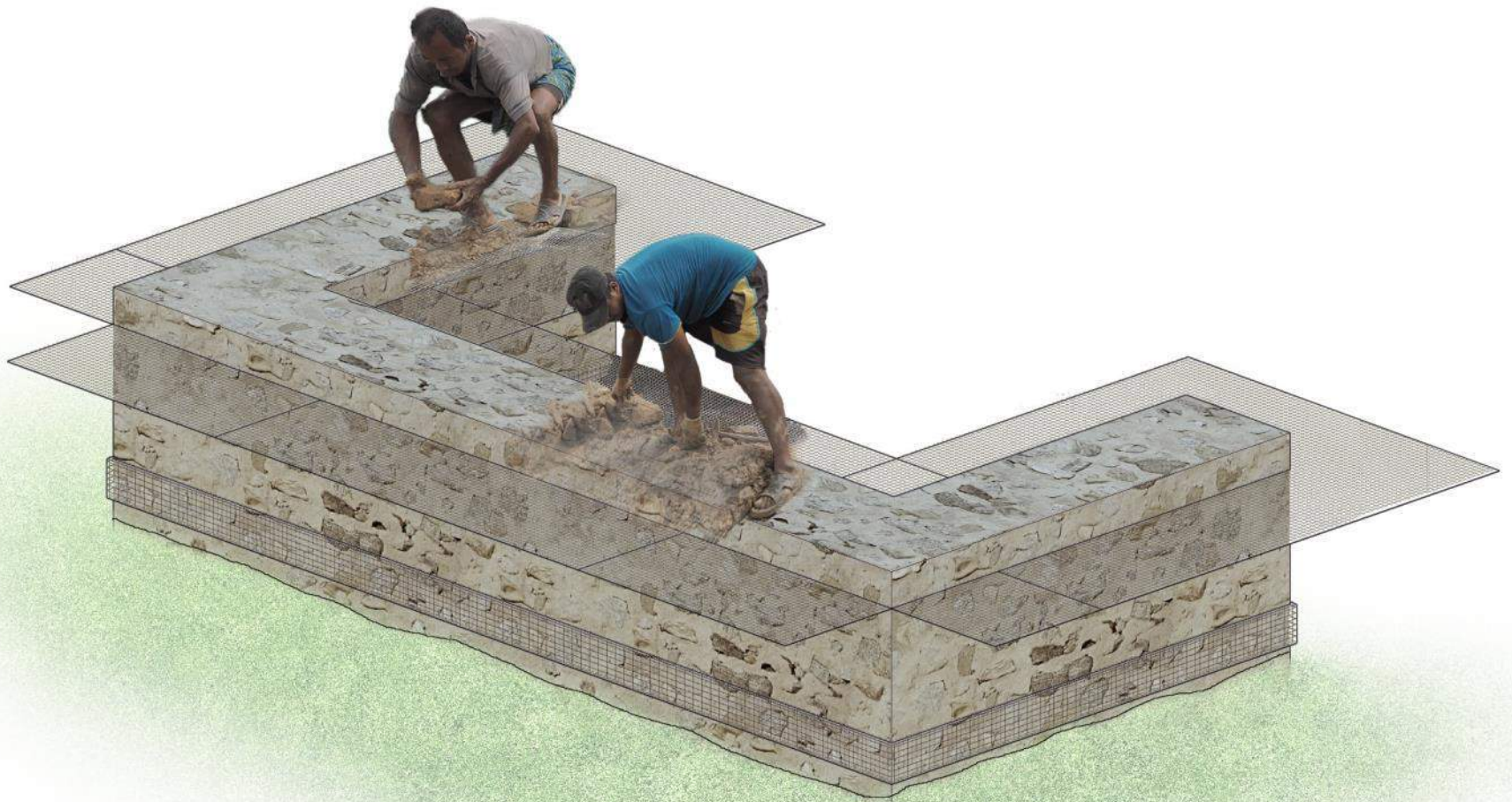




















Sunar Family Farm, 2015



NEPAL SCHOOL PROJECTS

Banakhu, Khavre District



NEPAL SCHOOL PROJECTS

Fran Lloyd, E.D.

Banakhu School, Khavre District

















CHUPAR SCHOOL

John Vavruska











Namaste



www.conservationtech.com

www.traditional-is-modern.net

